Servo and Display Controller Power Budget

Team Number: 102
Project Name: **Modular Spider Leg**
Name **Akshan Bhelkar**

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity Nano	DM164150	3.3-5.5 V board input	1	100	100	mA
	2.42-inch 128x64 I2C OLED	SSD1309 module	3-5 V module input	1	100	100	mA
	Hitec HS-485HB servo	HS-485HB	4.8-6.0 V	1	1200	1200	mA
	Indicator LED	Standard LED	5 V rail with 330 ohm resistor	2	10	20	mA
						0	

+5V Digital Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity Nano	DM164150	5 V	1	100	100	mA
	2.42-inch I2C OLED	SSD1309 module	5 V	1	100	100	mA
	Indicator LEDs	Standard LED	5 V	2	10	20	mA
					Subtotal	220	mA
					Safety Margin	25%	
					Total Current Required on +5V Digital Rail	275	mA
Regulator	+5V Digital Regulator	LM7805	7-35 V input	1	1500	1500	mA
					Total Remaining Current Available on +5V Digital Rail	1225	mA

+5V Servo Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Hitec HS-485HB servo	HS-485HB	4.8-6.0 V	1	1200	1200	mA
						0	mA
						0	mA
					Subtotal	1200	mA
					Safety Margin	25%	
					Total Current Required on +5V Servo Rail	1500	mA
Regulator	+5V Servo Regulator	LM7805	7-35 V input	1	1500	1500	mA
					Total Remaining Current Available on +5V Servo Rail	0	mA

Additional Power Rail - Not Used	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	No additional regulated rail is used in this subsystem.					0	mA

External Power Source	Component Name	Part Number	Supply Voltage Range	Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Power Source Selection	9 V DC Wall Adapter	T.B.D	100-240 VAC input	9 V DC	2000	2000	mA

Power Rails Connected to External Power Source	+5V Digital Regulator	LM7805	9 V input	1	275	275	mA
	+5V Servo Regulator	LM7805	9 V input	1	1500	1500	mA
						0	
					Total Remaining Current Available on External Power Source	225	mA

External Power Source 2 - Not Used

The subsystem uses one external 9 V DC wall adapter.

Battery Life - Not Applicable

This subsystem is powered by a wall adapter and does not use a battery, so battery life calculations are not applicable here.

Notes:

The HS-485HB worst case stall current is 1,200 mA. With a 25% safety margin, the servo rail requires 1,500 mA, equal to the assumed LM7805 maximum rating.

The 9 V, 2 A adapter is has extra available current of 225mA after safety margins. Leaving very little reserve but can icrease depending on the adapter used with potentialy higher amp rating.

At a 1.2 A stall load, the servo regulator may dissipate approximately (9 V - 5 V) x 1.2 A = 4.8 W. A bigger heatsink would be required.